

# Turnkey Full-Stack vs. Headless AI Engine

MoveDigital's agent platform was built with a strict decoupled boundary. Whether MyInstantAI deploys the **turnkey Next.js 15 app** on Day 1 or connects an in-house frontend to our **Headless AI Engine & Client SDK**, all core platform capabilities, multi-model resilience, and 99.9% API uptime SLAs remain 100% identical.

- Dual Integration Models
- Decoupled Architecture
- Enterprise 99.9% SLA

## 01 Architectural Integration Matrix

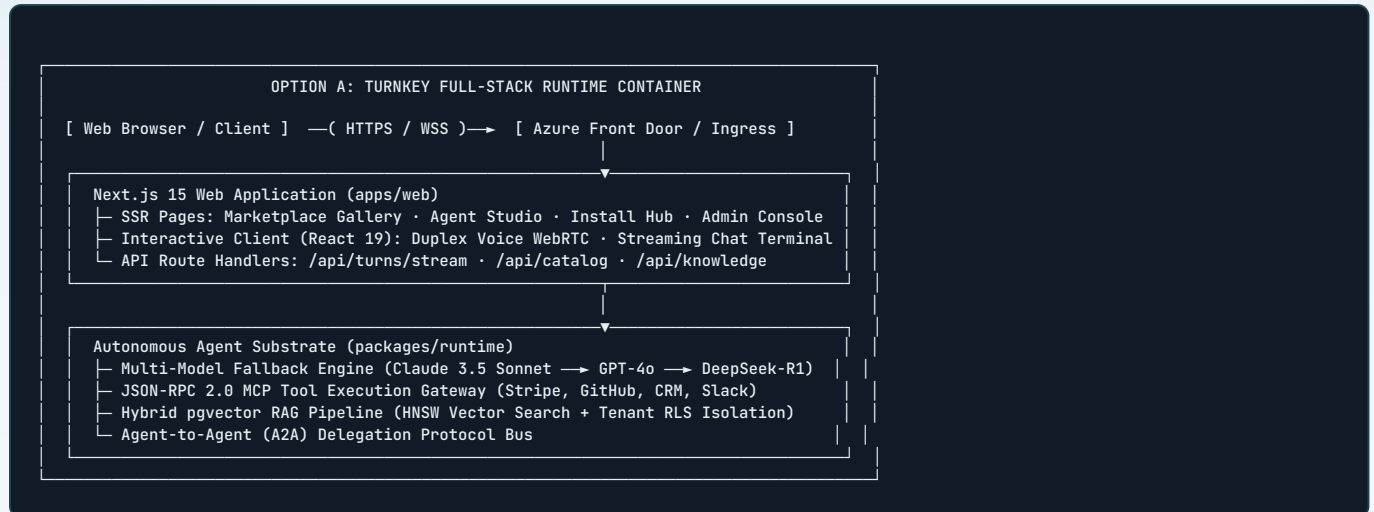
| DIMENSION              | OPTION A: TURNKEY FULL-STACK NEXT.JS 15                                                    | OPTION B: HEADLESS AI ENGINE & CLIENT SDK                                                        |
|------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Primary Scope          | End-to-end container (Next.js 15, React 19 UI, Studio, Hub + AI Engine)                    | Decoupled AI Engine microservice + TypeScript/Python SDK + OpenAPI REST/SSE                      |
| Time to Market         | <div>DAY 1 IMMEDIATE</div> — Zero frontend code required                                   | <div>3 TO 6 WEEKS</div> — Partner builds bespoke UI screens                                      |
| Frontend Ownership     | MoveDigital maintains web components, screens & responsive layouts                         | MyInstantAI internal engineering team designs & owns 100% of UI/UX                               |
| Integration Surface    | Branded CSS theme tokens, logo assets, OIDC auth config, domain DNS                        | Client SDK ( <code>@miai/client-sdk</code> ), <code>/api/turns/stream</code> (SSE), REST catalog |
| 500-Agent Catalog      | Rendered in pre-built marketplace gallery & configuration studio                           | Streamed via JSON catalog endpoints with typed tool parameters                                   |
| MoveDigital Core       | Application software, UI components, AI Engine, pgvector RAG, Multi-Model Router & App SLA | AI Engine microservice, Multi-Model Router, RAG, MCP Gateway, SDK Maintenance, 99.9% API SLA     |
| Cloud Hosting & Infra  | <b>MyInstantAI Azure Subscription</b> (Partner owns cloud spend & tenant)                  | <b>MyInstantAI Azure Subscription</b> (Partner owns cloud spend & tenant)                        |
| Identity & Auth (OIDC) | <b>MyInstantAI IdP (OIDC / Entra ID)</b> (App federates with partner IdP)                  | <b>MyInstantAI IdP (OIDC / JWT)</b> (Partner passes auth token to Client SDK)                    |
| Support & SLA          | Full Application & AI Runtime Maintenance SLA                                              | 99.9% AI Engine, API Gateway & Tool Execution SLA                                                |
| Deployment Footprint   | Single Container App (Web UI + Autonomous Runtime)                                         | Decoupled Microservice (API / SDK Substrate)                                                     |

## 02 Core Architectural Recommendation

**The Staged Onboarding Model:** Launch **Option A on Day 1** for immediate operational deployment, while MyInstantAI frontend engineers build against the MoveDigital Client SDK in parallel (**Option B**). When the custom UI is completed in Week 5, cutover requires zero database migrations or backend refactoring.

## 83 Option A: Turnkey Full-Stack Deployment

Deploy the complete enterprise application as a single Azure Container App. MoveDigital manages both the frontend presentation tier and the agent execution runtime.



### Included Out-of-the-Box Modules

#### 1. Agent Marketplace & Storefront

Categorized directory of 500 domain agents, fuzzy search, capability filters, price tiers, and one-click sandbox testing.

#### 2. Agent Studio & Customizer

Prompt configuration studio, knowledge file upload manager, dynamic temperature tuning, and tool permission toggles.

#### 3. Duplex Voice Interface

Low-latency conversational voice engine with real-time waveform visualization, interruptibility, and audio playback.

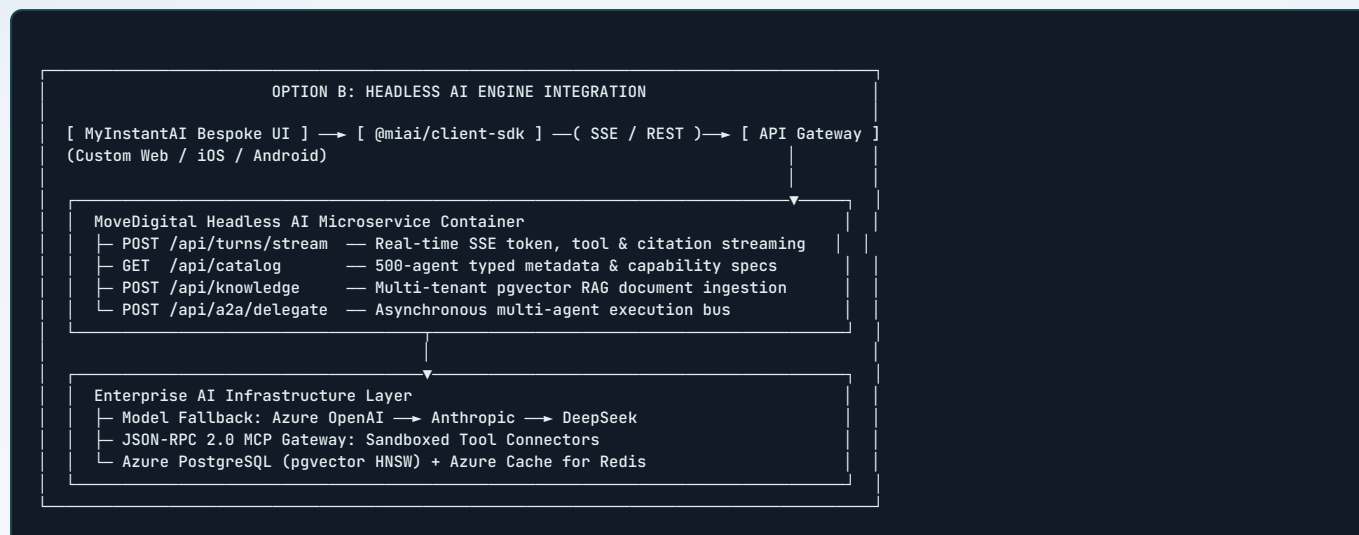
#### 4. Integration & Install Hub

OAuth connector setup for Slack, Microsoft Teams, WhatsApp Web, and custom webhooks with tenant-level credential storage.

**When to Choose Option A:** Recommended if MyInstantAI wants immediate deployment on Day 1, lacks frontend bandwidth, or wants MoveDigital to shoulder the complete UI maintenance and browser-compatibility surface.

## 84 Option B: Headless AI Engine & Client SDK

MyInstantAI designs, builds, and hosts the frontend UI across web, mobile, and enterprise portals. MoveDigital delivers a high-throughput, low-latency AI microservice via Client SDK and REST/SSE APIs.



### Client SDK Developer Snippet (TypeScript)

```
import { MoveDigitalAgentClient } from "@miai/client-sdk";

const client = new MoveDigitalAgentClient({
  baseUrl: "https://ai-engine.myinstantai.digital",
  apiKey: userAuthToken,
  tenantId: "tenant_enterprise_01"
});

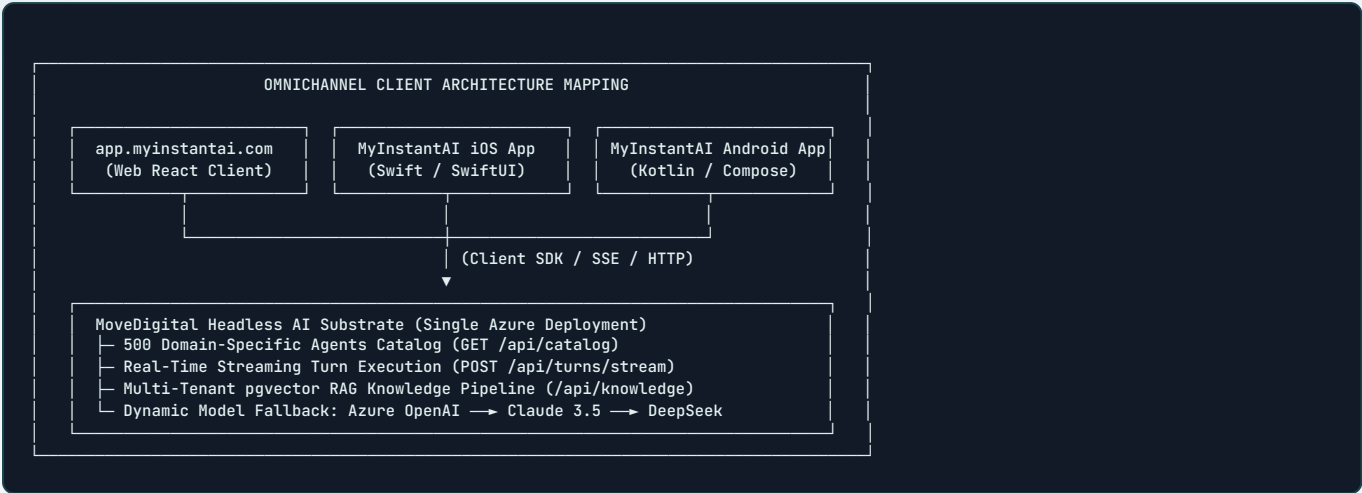
// Stream tokens, tool calls, and citations directly into MyInstantAI custom state
const stream = await client.turns.stream({
  agentId: "financial-auditor-pro",
  message: "Verify compliance with IFRS 16 lease liability rules.",
  sessionId: "session_abc123"
});

for await (const event of stream) {
  if (event.type === "chunk") appendTokenToUI(event.data.delta);
  if (event.type === "tool.call") renderToolSpinner(event.data.tool);
  if (event.type === "grounding.citation") renderBadge(event.data.source);
}
```

**Headless Engine Core Surface:** Exposes OpenAPI 3.1 REST endpoints and Server-Sent Events (SSE) streaming token deltas, tool executions, and vector citations with zero coupling to React or frontend CSS.

85 Direct Platform Mapping: app.myinstantai.com, iOS & Android

A technical review of MyInstantAI's live production web platform ([app.myinstantai.com](https://app.myinstantai.com)) and native mobile apps (iOS & Android) demonstrates why Option B is the ideal architectural fit.



1. Web Portal Integration Blueprint ([app.myinstantai.com](https://app.myinstantai.com))

### Sidebar Navigation Alignment

Add an "AI Agents" entry directly under the CORE section (Home, Ask AI, AI Agents, Search, History). Clicking it opens the 500-agent catalog grid.

### Interactive Canvas Transformation

Selecting an agent shifts the workspace from generic model chatting (**Single / Compare / Blend**) into **Autonomous Agent Execution** with live tool status badges.

### 2. Native Token & e-PIN Balance Ledger Alignment:

[app.myinstantai.com](https://app.myinstantai.com) already features a prominent **Token Balance Widget** (e.g. 11,716 FOREVER) backed by **Redeem e-PIN** and subscriptions. When a turn completes, MoveDigital's engine emits a typed **turn.complete** event with exact prompt and completion token counts. MyInstantAI's billing engine simply deducts tokens from the user's existing ledger with zero disruption to their monetization model.

### 3. Native Mobile Parity (iOS & Android):

Because MoveDigital's engine uses standard SSE streams, MyInstantAI's iOS (Swift) and Android (Kotlin) apps consume the exact same agent catalog, RAG citations, and tools without developing separate mobile backends.

86 Architectural Value & RACI Division

**Architectural Value Boundary:** The core technical moat resides in model resilience, pgvector hybrid search, MCP sandboxing, and prompt engineering—not frontend markup. Building an autonomous AI engine from scratch requires specialized AI/ML infrastructure engineering; MoveDigital provides this substrate with battle-tested reliability and an enterprise 99.9% API SLA.

| CAPABILITY / RESPONSIBILITY AREA                     | OPTION A: TURNKEY FULL-STACK                    | OPTION B: HEADLESS ENGINE                       |
|------------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| UI/UX Design, Components & Layouts                   | MoveDigital                                     | MyInstantAI In-House Team                       |
| Cross-Browser QA & Mobile Responsive Polish          | MoveDigital                                     | MyInstantAI In-House Team                       |
| Agent System Prompts & 500-Catalog IP                | MoveDigital                                     | MoveDigital                                     |
| Multi-Model Failover & Token Optimization            | MoveDigital                                     | MoveDigital                                     |
| pgvector RAG Chunking & Hybrid Search                | MoveDigital                                     | MoveDigital                                     |
| Model Context Protocol (MCP) Tool Sandboxing         | MoveDigital                                     | MoveDigital                                     |
| A2A Multi-Agent Orchestration Protocol               | MoveDigital                                     | MoveDigital                                     |
| Azure Cloud Subscription & Infra Ownership           | MyInstantAI (Owns Azure tenant & cloud spend)   | MyInstantAI (Owns Azure tenant & cloud spend)   |
| Identity Provider & User Directory (OIDC / Entra ID) | MyInstantAI (Owns IdP, users, roles & SSO)      | MyInstantAI (Owns IdP, users, roles & SSO)      |
| Cloud Provisioning & Network Security                | MyInstantAI (with MoveDigital config templates) | MyInstantAI (with MoveDigital config templates) |
| Application Container & Code Updates                 | MoveDigital                                     | MoveDigital                                     |
| 24/7 Engine Health & 99.9% Uptime SLA                | MoveDigital                                     | MoveDigital                                     |

87 Recommended Action Plan

|                                                                                                                               |                                                                                                                                           |                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1: Day 1–15</b><br>Deploy Option A on Azure. Brand with MyInstantAI logos and begin customer onboarding immediately. | <b>Phase 2: Day 15–40</b><br>MyInstantAI frontend team consumes Client SDK in parallel, building bespoke portal against live staging API. | <b>Phase 3: Day 45+</b><br>Cutover traffic to bespoke UI. Zero backend migration required; all agent memories and RAG vectors persist. |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|